

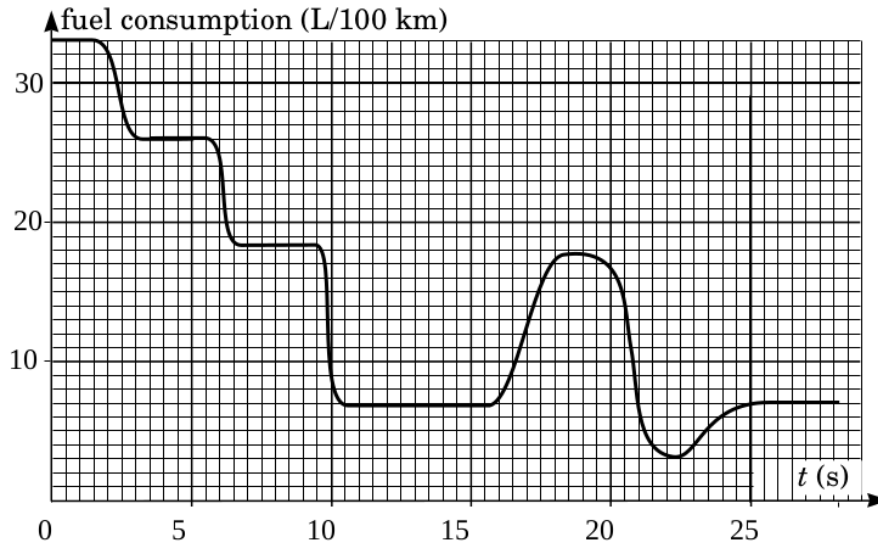
Mechanics – 13/15.03.2026

Some of these are quite difficult, so don't hesitate to ask for hints. For the Friday class, please send your work by **23:00 on 12.03 (Thursday)**. You should cover at least half the problem set.

Kinematics

Problem 1. Conical room. See [NBPhO 2020, Problem 3](#).

Problem 2. Fuel consumption. See [NBPhO 2016, Problem 1](#). You'll need to make measurements on the graph below:

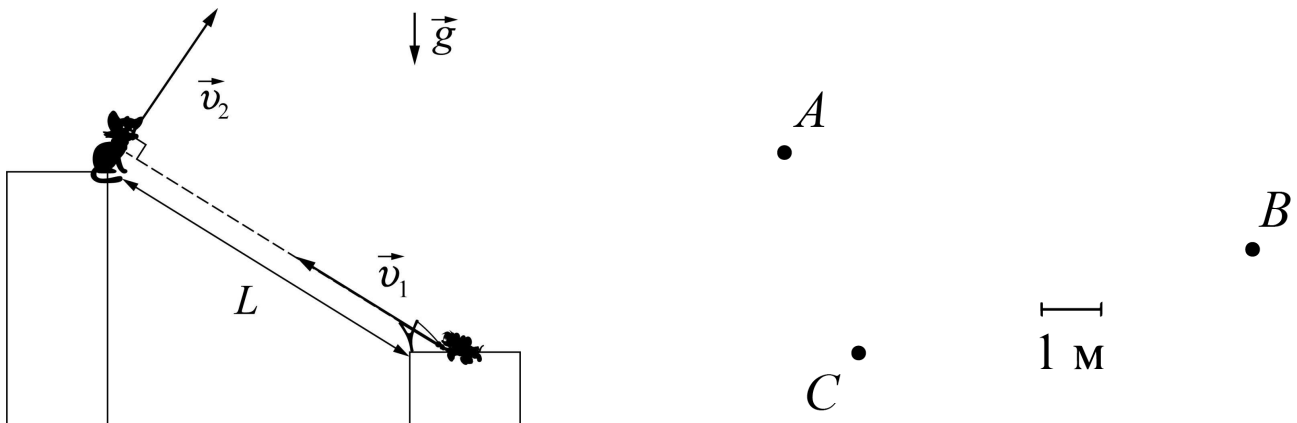


Problem 3. Cat and mice. Seeing Leopold standing on the edge of the opposite bank of a ravine, the mice threw a stone directly towards him with a speed v_1 . Leopold noticed this and, at the moment of the throw, jumped with a speed v_2 in the plane of the stone's trajectory, perpendicular to the line joining him and the mice (see the figure). When the distance between Leopold and the stone was minimal, their velocities again became perpendicular.

(a) Find the maximum possible initial distance $L_{\max}$ between Leopold and the mice.

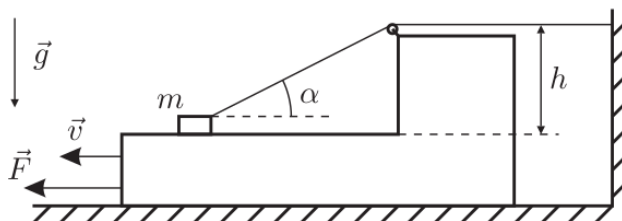
Now we work in the case where the initial distance is precisely $L_{\max}$. At the instant when the distance between Leopold and the stone was minimal, someone took a photo of Leopold (A), the mice (B), and the stone (C). The orientation of the photo is unknown. Note the scale on the photo.

(b) Find v_1 and v_2 . The acceleration due to gravity is $g = 9.8 \text{ m/s}^2$. (Russia 2020)



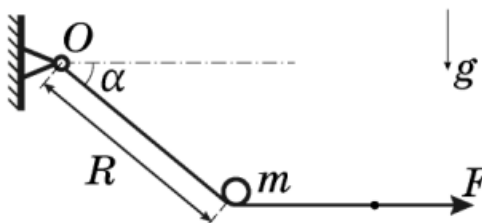
Newton's 2nd law

Problem 4. Platform. On top a platform with a rectangular step of height h there is a small body of mass m . It's attached to one end of a light inextensible thread that goes over an ideal pulley placed on top of the step. The other end of the thread is attached to a vertical wall, such that the thread is horizontal between the pulley and the wall. The platform is pushed away from the wall with a constant velocity v . For this to happen, what is the horizontal force F that we'll need to apply to the platform at the instant when the thread makes an angle α with the horizon? The coefficient of friction between the platform and the small body is μ , while the floor is smooth. Assume that the body, the platform, and the floor always remain in contact. (Russia 2012)



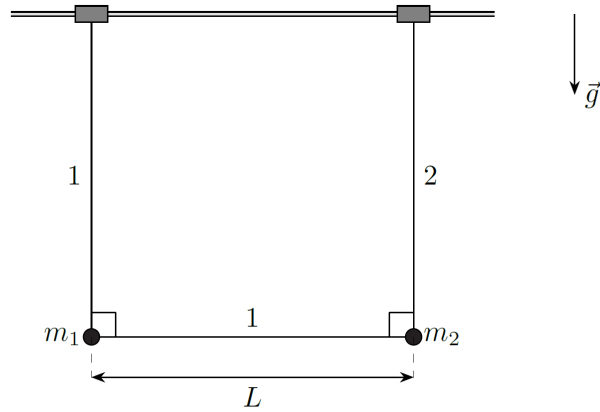
Conservation laws

Problem 5. Cylinder and band. A thin, light, and inextensible band is attached to the wall at point O . On the band, we hold tight a small cylinder of mass m , such that the inclined part of the band has length R and makes an angle α with the horizon. We let the cylinder go and we begin to apply a constant horizontal force F to the free end of the band. The cylinder starts moving. Find the speed of the cylinder at the instant when it detaches from the band. Neglect all friction. The acceleration due to gravity is g . (Russia 2018 Regionals)



Problem 6. Collision on a spoke. Two identical small, massless, short cylinders slide without friction along a long, fixed horizontal spoke. The cylinders are connected to weightless, inextensible threads 1 and 2, with the length of thread 1 being greater than the length of thread 2 by L . A small bead of mass m_1 can move along thread 1 without friction, and a small block of mass m_2 is attached to the ends of both threads. Initially, the system is held at rest such that the spoke and both threads are in the same vertical plane, the bead and block are located on the same horizontal plane, the bead is located on the same vertical line as the cylinder attached to thread 1, and the block is located on the same vertical line as the cylinder attached to thread 2. The threads are slightly taut. The acceleration due to gravity is g . Assume the cylinders are larger than the bead and the block. Neglect air resistance. The system is let go.

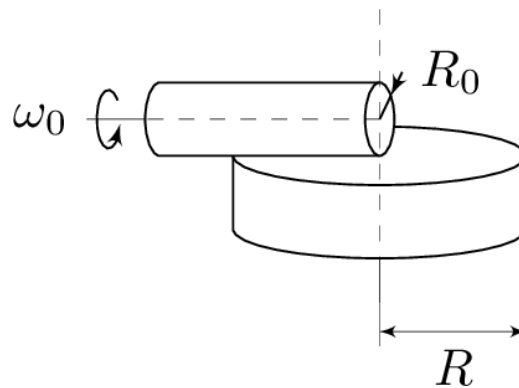
- (a) Determine the velocities v_1 and v_2 of the bead the block, respectively, right before the cylinders collide.
- (b) Find the tension forces T_1 and T_2 of threads 1 and 2, respectively, right before the cylinders collide. (Russia 2024)



Rotation

Problem 7. Friction drive. A long cylindrical roller of radius R_0 , rotating around its axis with an angular velocity ω_0 , is uniformly pressed against a freely (i.e. without friction in the axis) rotating disk of radius R . The line of contact between the disk and the roller coincides with the radius of the disk (see the figure).

- Find the steady-state angular velocity ω_μ of the disk's rotation if the friction between the roller and the disk is dry.
- Find the steady-state angular velocity ω_η of the disk's rotation if the friction is viscous. Assume that the magnitude of the viscous friction force per unit contact length is proportional to the relative velocity of the contacting surfaces of the roller and disk. Find $k = \omega_\eta/\omega_\mu$. (Russia 2016)



Problem 8. Car. See [NBPhO 2006, Problem 6](#). You'll need to draw here. I've copied the diagram for convenience. You can either print it out or screenshot it, import it in e.g. GeoGebra, and measure there.

